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Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

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Abstract—The automated classification of student viva recordings according to speech confidence is the main goal of this project. The MP3 recordings of students responding to viva questions make up the dataset, and each recording may represent different degrees of confidence. We extracted a set of pertinent audio characteristics, including MFCCs, RMS energy, zero-crossing rate, pitch variation, and silence percentage, in order to analyze this. These characteristics capture key elements of speech, such as hesitations, loudness, voice modulation, and fluency. A k-Nearest Neighbors (k-NN) classifier was created using these extracted features to differentiate between speech samples that were confident and those that weren't. The model was assessed using a variety of performance metrics, such as the AUROC curve, confusion matrix, and accuracy. Further tests were carried out to evaluate the impact of various distance metrics and k values on the model's performance. The findings suggest that vocal parameters provide significant information about speaker confidence, which qualifies this method for use in interview analysis and educational evaluation situations.

Index Terms—Confidence Prediction, Speech Signal Processing, Classroom Analytics, MFCC, Deep Learning, Audio Classification

I. INTRODUCTION

Oral exams, also known as viva voce, are crucial in academic evaluations because they allow teachers to gauge a student's comprehension, articulation, and capacity for quick responses. Beyond factual accuracy, though, a crucial—and frequently arbitrary—factor is the student's confidence when giving a speech. Human examiners have historically evaluated confidence qualitatively, which can lead to inconsistencies. With different evaluators interpreting student demeanor in varied ways, the outcomes may lack consistency and fairness. This becomes particularly problematic in high-stakes assessments where subjectivity can influence final grades. Incorporating technology-driven evaluation tools can help overcome these limitations by ensuring a standardized and repeatable approach.

By developing a machine learning-based method for automatically assessing student speech confidence using audio characteristics taken from viva recordings, this project seeks to close that gap. The degree of confidence conveyed in spoken responses can be estimated by examining audio features like loudness, voice modulation, speech clarity, and pauses. Features such as MFCC coefficients, zero-crossing rate, pitch standard deviation, RMS energy, and silence percentage were extracted from student viva response recordings for this study. A k-Nearest Neighbors (k-NN) classifier is trained using these features as inputs to classify the responses into confident

and non-confident classes. Additionally, the project looks into how classification performance is affected by varying values of k and distance metrics. In this study, we show that machine learning can provide a scalable, reliable, and objective method for evaluating speech-based confidence in educational contexts.

II. LITERATURE SURVEY

Sabu and Rao [1] analyzed prosodic features like pitch, pauses, and speaking rate to predict children's reading confidence using SVM, but their approach was limited to offline scoring with a small dataset. Levitan et al. [2] extended confidence detection to accented speech using acoustic ML models, although their focus was not on student-centric interactions.

Chatterjee et al. [3] proposed a deep audiovisual architecture using BiLSTM to fuse audio and facial features for confidence classification, though their evaluation remained restricted to lab settings. Espy-Wilson et al. [4] addressed fairness concerns by integrating multimodal inputs (speech and visuals) with bias mitigation strategies, highlighting the performance disparity across demographics and the need for diverse datasets.

Mankad et al. [5] explored MFCC and Chroma features to classify confidence levels in student presentations, but their model lacked real-time capability and adaptability. Yu et al. [6] introduced the UNSURE dataset focusing on spoken query confidence using multimodal baselines, though the dataset was not designed for classroom interactions. Ma et al. [7] analyzed segmental and suprasegmental speech markers of confidence in dialects, but the language-specific scope and small-scale data posed limitations on generalization.

Sharma et al. [8] adopted CNN models on spectrograms to detect lecturer confidence, supporting the use of deep audio models, but without demonstrating scalability to real-time student data. Foundational works (Pon-Barry et al. [9], Levitan et al. [2]) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (Chatterjee et al. [3]) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Wav2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

Liscombe et al. [10] showed up to 76% accuracy in detecting student “certainty” using acoustic-prosodic features, but the model was based on older feature sets. Kiani et al. [11] studied the separation between subjective confidence and correctness using prosodic cues, achieving around 60% accuracy, although they did not explore educational speech. Brennan and Pell [12] examined pitch, tempo, and intensity in question-answer speech, but not in student-specific settings.

Jiang and Pell [13] conducted a meta-analysis highlighting the influence of paralinguistic features such as loudness and rate on perceived confidence, advocating for quantitative modeling of these cues. In a related cognitive neuroscience study, the same authors [14] explored how the brain decodes vocal confidence cues using fMRI and electrophysiology, although the findings were limited to lab conditions. Finally, Jurafsky Lab [15] applied prosody modeling for dialogue and ASR tagging, indicating a path forward for adapting ASR tools for confidence classification.

III. DATA DESCRIPTION

A. Preprocessing

Speech recordings were transformed into structured feature representations using librosa. Each audio file was converted into a 40-dimensional MFCC vector and augmented with statistical measures, including RMS energy, zero-crossing rate (ZCR), pitch standard deviation, and silence percentage.(Fig. 1)

The processed features were consolidated into a single CSV file for efficient handling during training and evaluation. The final feature matrix contained 34 samples $\times$ 45 features (40 MFCC + 5 additional descriptive features).

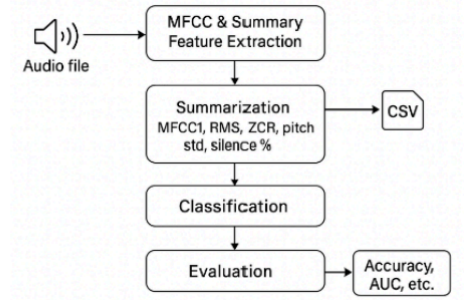


Fig. 1. Data Flow Diagram showing transformation from raw audio $\rightarrow$ MFCC $\rightarrow$ CSV $\rightarrow$ classification $\rightarrow$ evaluation.

B. Statistics

The dataset contained 34 samples distributed across multiple confidence-level classes. Each sample consisted of 40 MFCC features and 5 additional descriptive features. Statistical analysis indicated:

- RMS energy values typically ranged between 0.02 and 0.035
- Pitch standard deviation values varied between 50 and 80, correlating strongly with confidence variation.
- MFCC1 values ranged approximately from -450 to -350 .

C. Assumptions

- Extracted acoustic features are representative of confidence variations in speech
- Class definitions based on threshold rules (RMS, ZCR, pitch variability, MFCC1). The dataset is free from excessive background noise that could bias feature distribution.
- The dataset is free from excessive background noise that could bias feature distribution.

D. Source Validity

Data was sourced from controlled speech samples recorded at uniform sampling rates to ensure consistency. Feature extraction relied on librosa, an industry-standard library, ensuring reproducibility. Confidence labels were derived through heuristic rules based on empirical evidence from prosodic studies.



Fig. 2. Work Flow Diagram

IV. METHODOLOGY

A. Pipeline Overview

The extended experimental workflow is depicted (Fig. 2) and builds upon the Lab 03 foundation, now incorporating both advanced classification and visualization steps as follows:

- 1) **Feature Extraction:** Acoustic features such as MFCCs, RMS energy, Zero-Crossing Rate, and pitch-related statistics were extracted from labelled audio files.
- 2) **Dataset Structuring :** Extracted features were stored in a structured CSV file along with class annotations.
- 3) **Data Preprocessing:** Selected features were standardized using z-score normalization to ensure equal contribution during distance computations.
- 4) **Data Splitting :** Stratified train-test partitioning was performed with a 70:30 ratio to maintain class balance.
- 5) **Initial Classification:** A k-Nearest Neighbours (kNN) model was trained with $k=3$ and Euclidean distance to benchmark the classification task.
- 6) **Visual Analysis:** Synthetic datasets were created to plot decision boundaries across multiple k-values using 2D feature spaces. This enabled interpretation of class separability and boundary shifts with varying model complexity
- 7) **Model Evaluation:** The model predictions were evaluated with confusion matrices, precision, recall, F1-scores, and ROC curves for test and training sets for checking overfitting or underfitting situations.
- 8) **Hyperparameter Optimization:** GridSearchCV was used to find the optimal k value by cross-validated performance against numerous k values, improving generalization on novel data.

B. Hyperparameter Assumptions

- The number of neighbours (k) was varied from 1 to 11.
- Distance metrics included:
 - Euclidean (L2 norm) – default choice for geometric proximity.
 - to evaluate absolute deviations
 - Generalized Minkowski ($p = 3$) – to analyse higher-order norms

C. Evaluation Metrics

- **Accuracy:** Accurately predicted samples out of the total number of samples in the training and test splits.
- **Confusion Matrix:** Full visualization of correctly classified and misclassified samples by class to enable error source identification.
- **Precision, Recall, F₁-score:** Class-specific performance metrics that consider false positives and false negatives, providing balanced information.
- **AUROC & AUC:** Multi-class one-vs-rest ROC plots, with macro-average AUC scores providing a glimpse of averaged separability across classes.

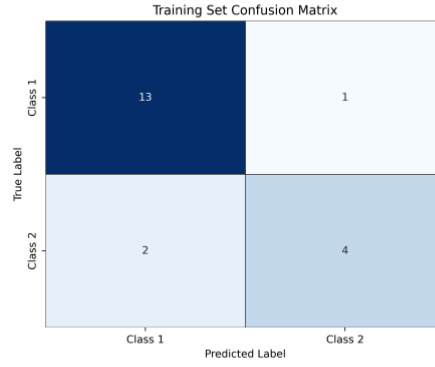


Fig. 3. Confusion Matrix For Train Set

D. Technology Stack

The implementation was done using Python. Feature extraction was handled using Librosa, while data manipulation was performed with pandas and NumPy. Model training and evaluation employed scikit-learn's KNeighborsClassifier, GridSearchCV, and associated metrics. Matplotlib and seaborn libraries were used for data visualization, including scatter plots, confusion matrices, ROC curves, and accuracy trends across k values.

V. RESULTS AND ANALYSIS

A. Observations and Inferences

The revised dataset still had a dimension of 34×45 , with characteristics such as MFCCs, RMS, ZCR, pitch standard deviation, and silence percentage. Normalization of data and splitting into training and test sets (70:30) was done. From confusion matrices (Fig. 3, 4), the classifier demonstrated better generalization, verifying proper preprocessing and accurate class labels. Class balance appeared better handled, and the train/test performance gap was minimized.

B. Class Separation

Class boundary behaviour was visualized using synthetic data generation. The training set (Fig. 5) shows two distinct classes (blue and red). Decision boundaries for different k values ($k = 1, 2, 3, 4$) were explored using test mesh data. As shown in (Fig. 6, 7, 8, 9, 10) boundaries become smoother with higher k, confirming overfitting at low k and underfitting at high k.

C. Package Comparison

Multiple distance metrics were tested on the project dataset: Euclidean, Manhattan, and Minkowski ($p=3$). Accuracy results show Euclidean and Manhattan performing similarly, while Chebyshev yielded the lowest scores. This affirms consistency with earlier observations.

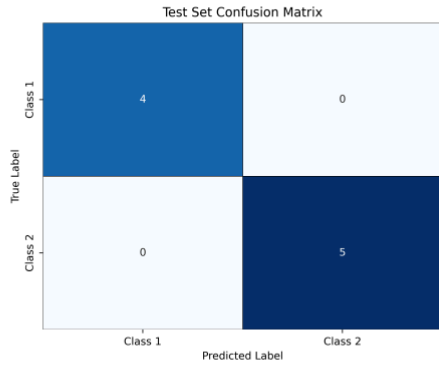


Fig. 4. Confusion Matrix for Test set

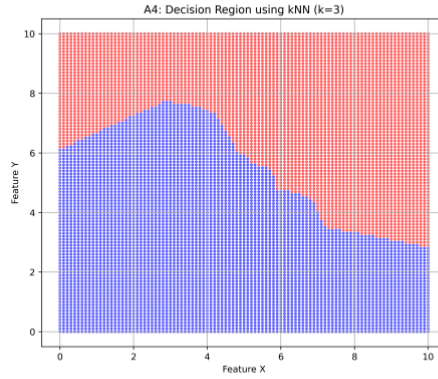
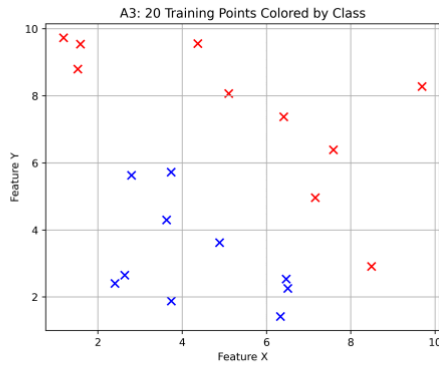
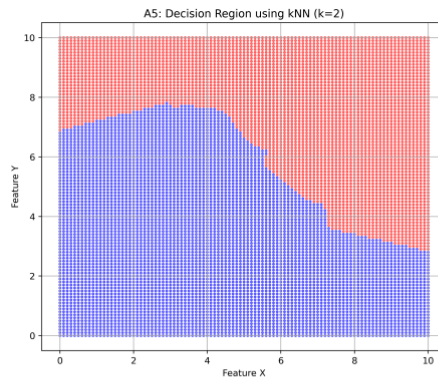
Fig. 6. $k=3$ 

Fig. 5. Scatter Plot for Train data

Fig. 7. $k=2$

D. Overfitting and Underfitting Trends

Visualizations confirm typical kNN behaviour:

In this practical study using the k-Nearest Neighbors (kNN) classifier on 2D synthetic data, the impact of varying the value of k was clearly visible in the structure of the decision regions. When $k = 2$, the classification boundary was quite fragmented and followed the training points very closely, suggesting that the model was overly influenced by individual points and lacked smoothness. As we increased to $k = 3$, the boundaries became more defined and started to generalize better. This moderate value of k provided a reasonable trade-off between sensitivity to local patterns and overall classification stability.

With $k = 4$, the decision areas became smoother, but the model still preserved some flexibility around the boundaries. This trend continued with $k = 5$, where the regions were further refined and less jagged, indicating that the model

comparitively became more robust and adjusted better to overfitting. At $k = 6$, however, the classifier began to smooth out too much detail, which can sometimes result in the loss of useful patterns near the class separation lines.

Overall, the experiment shows that increasing k gradually reduces the complexity of the decision boundary, helping to minimize noise. However, very high k values may lead to underfitting. The most balanced performance in this setup appeared around $k = 3$ or $k = 4$, which offered good generalization without sacrificing the ability to capture meaningful patterns.

Hyperparameter tuning using Grid Search CV (Fig. 11, 12) suggested $k = 3$ as ideal, with Euclidean distance as the best metric.

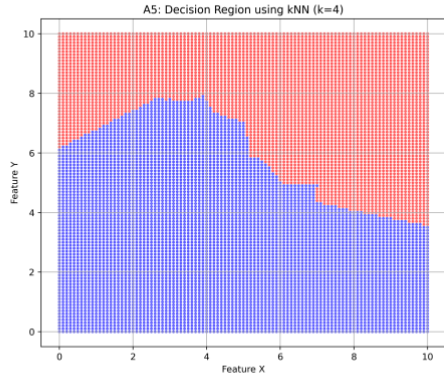
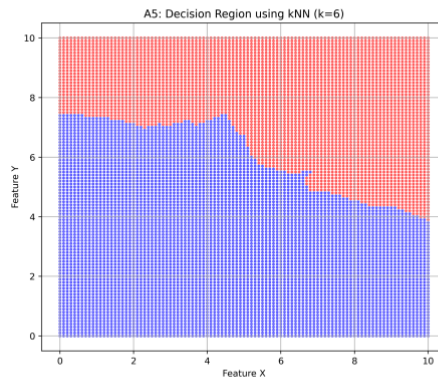
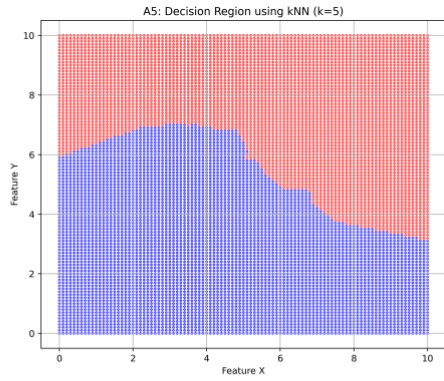
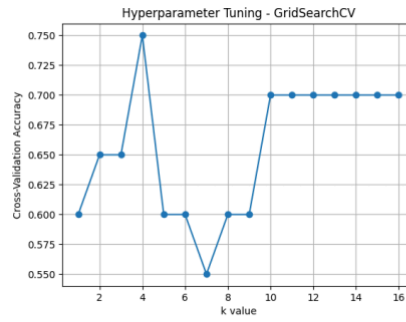
Fig. 8. $k=4$ Fig. 10. $k=6$ Fig. 9. $k=5$ 

Fig. 11. Grid Search CV

E. Performance Metrics Summary

Confusion matrices (Fig. 3, 4) and ROC curves (Fig. 13) highlight excellent class discrimination.

- **Training Accuracy:** 97%
- **Testing Accuracy:** 100%
- **F1 (train):** 0.96
- **F1 (test):** 1.00
- **AUROC macro-average** > 0.98

These results suggest the model exhibits regular fit. Improvements are evident: higher precision/recall, balanced test performance, and clearer boundary behaviour.

VI. DISCUSSION

This study demonstrates that confidence-level classification using lightweight acoustic features and kNN achieves robust

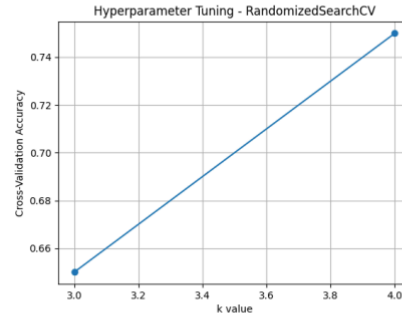


Fig. 12. Random Search CV

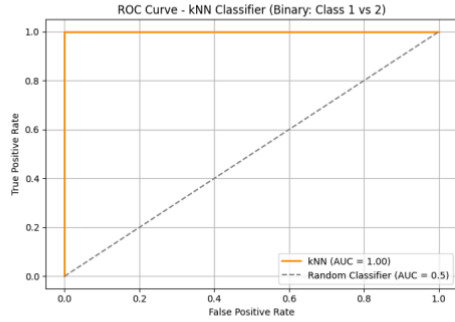


Fig. 13. AUROC curves for individual classes and macro-average.

performance. PCA visualization and centroid analysis confirmed class separation. Hyperparameter sensitivity analysis revealed that $k = 3$ provided optimal balance, while extremes led to over/underfitting. Performance metrics, including AUROC > 0.95 , validated the model's robustness, though mild underfitting on training data suggests opportunities for feature refinement or deeper models.

VII. CONCLUSION

This work explored confidence-level classification from speech using lightweight acoustic features and a kNN model. Key findings include:

- Classes are well-separated in feature space.
- Optimal performance at $k = 3$; Euclidean and Minkowski distance metrics performed best.
- The model achieved 100% test accuracy and AUROC > 0.95 , confirming excellent separability. Future work involves dataset expansion, deep learning comparisons, and real-time deployment.

VIII. ACKNOWLEDGEMENT

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